

01-HelloWorld

August 3, 2022

1 Example Python Progam

```
# Iterative Binary Search Function
# It returns index of x in given array arr if present,
# else returns -1
def binary_search(arr, x):
    low = 0
    high = len(arr) - 1
    mid = 0

    while low <= high:
        mid = (high + low) // 2

        # If x is greater, ignore left half
        if arr[mid] < x:
            low = mid + 1

        # If x is smaller, ignore right half
        elif arr[mid] > x:
            high = mid - 1

        # means x is present at mid
        else:
            return mid

    # If we reach here, then the element was not present
    return -1


# Test array
arr = [ 2, 3, 4, 10, 40 ]
x = 10


# Function call
result = binary_search(arr, x)


if result != -1:
    print("Element is present at index", str(result))
```

```
else:
    print("Element is not present in array")
```

2 Example C Program

```
#include <stdio.h>

// A iterative binary search function. It returns location of x in
// given array arr[l..r] if present, otherwise -1
int binarySearch(int arr[], int l, int r, int x)
{
    while (l <= r)
    {
        int m = l + (r-l)/2;

        // Check if x is present at mid
        if (arr[m] == x)
            return m;

        // If x greater, ignore left half
        if (arr[m] < x)
            l = m + 1;

        // If x is smaller, ignore right half
        else
            r = m - 1;
    }

    // if we reach here, then element was not present
    return -1;
}

int main(void)
{
    int arr[] = {2, 3, 4, 10, 40};
    int n = sizeof(arr)/ sizeof(arr[0]);
    int x = 10;
    int result = binarySearch(arr, 0, n-1, x);
    (result == -1)? printf("Element is not present in array")
                  : printf("Element is present at index %d", result);

    return 0;
}
```

3 Obligatory “Hello World”

```
[1]: print("Hello World")
```

Hello World

4 Python 2 vs Python 3

```
[2]: # python 2  
# >>> print 5/2  
# >>> 2  
  
# python 3  
print(5/2)
```

2.5